

External Natural Convection

Natural Convection – Lesson 3



/ Intro to External Natural Convection

- When studying external natural convection, the analysis of hot or cold bodies immersed in a static fluid is the primary objective. The aim is to estimate the **heat transfer due to external natural convection**.
- In this lesson, we will analyze simple geometries such as vertical, oblique and horizontal plates; cylinders; and spheres.
- Real world applications can in many cases be approximated by such canonical geometries, e. g., for the purpose of preliminary analysis.
- Relations for cylinders can be useful for applications such as heating or cooling pipes. In these cases, it is important to be able to estimate the **potential heat losses** due to the external environment that would cause the fluid to cool down or heat up while its moving through the pipes.
- It can also be useful to understand if pipe insulation would perform adequately.



Pipe insulation



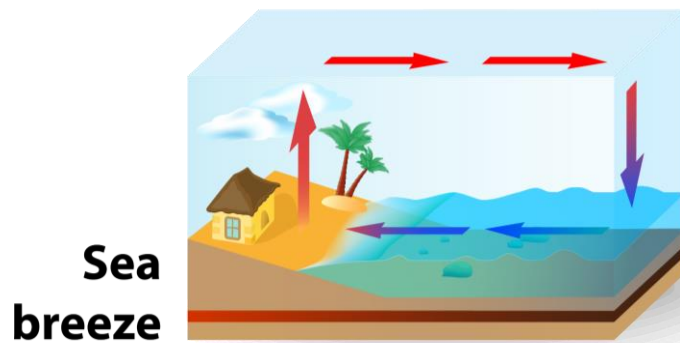
Radiator – Steam heating

/ Introduction to External Natural Convection (cont.)

- A straight surface can be analyzed using the relations for plates. Building walls or windows are good examples for such approximations when there is no wind.
- Some particle accelerators use natural air convection to cool their electromagnets. The coils are tightly packed and can be approximated by a simple plate.
- Land and water surfaces (like lakes) exchange heat with the air, creating convective patterns that affect atmospheric circulation.
- External natural convection can also be observed in everyday objects. A hot coffee mug cools down by releasing some of its heat into the surrounding air through natural convection.



Particle accelerator equipment



Atmospheric circulation

External Natural Convection Analysis

- The temperature difference between the fluid and the solid creates density gradients in the fluid that dictate whether the fluid would flow up or down the body along the direction of the gravitational field.
- The heat transfer analysis relies on estimating the **average Nusselt number** using **correlations**, in order to find the heat transfer coefficient and calculate the heat transfer due to convection. The correlations can be generalized using this form:

$$\overline{Nu}_L = \frac{\bar{h}L}{k} = C Ra_L^n$$

where

$$Ra_L = Gr_L Pr = \frac{g\beta(T_s - T_\infty)L^3}{\nu\alpha}$$

- The Rayleigh number is based on the **characteristic length** L of the body analyzed. The fluid properties are estimated at the film temperature defined as:

$$T_f = \frac{(T_s + T_\infty)}{2}$$

- The exponent n in first expression on this page is typically $1/4$ for laminar flows and $1/3$ for turbulent flows. The latter immediately tells us that the heat transfer coefficient is independent on the body characteristic length.

Vertical Plate

- In the case of natural convection on a vertical plate we can find correlations for **laminar** and **turbulent** flows in the form presented in the previous slide:

Laminar

$$\overline{Nu}_L = 0.59 Ra_L^{1/4}$$

$$10^4 \lesssim Ra_L \lesssim 10^9$$

Turbulent

$$\overline{Nu}_L = 0.1 Ra_L^{1/3}$$

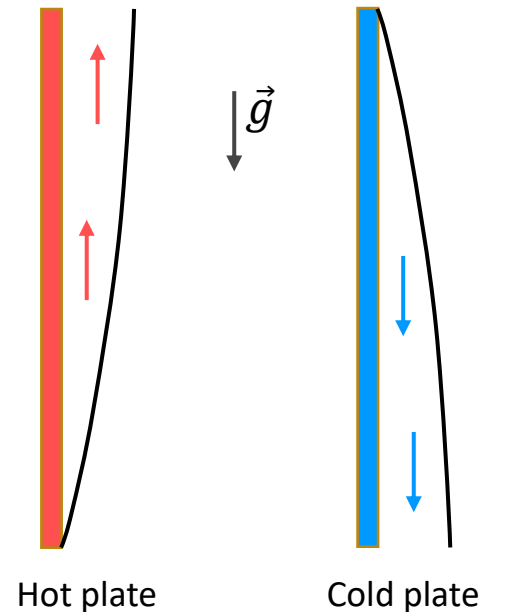
$$10^9 \lesssim Ra_L \lesssim 10^{13}$$

- However, these relations have a limited range of applicability and are connected to the fluid nature. The following correlation can be used for **any value** of Rayleigh number, instead:

$$\overline{Nu}_L = \left\{ 0.825 + \frac{0.387 Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right\}^2$$

- For laminar flow on vertical plates due to convection ($Ra_L \lesssim 10^9$) a similar correlation could be used if additional accuracy is needed:

$$\overline{Nu}_L = 0.68 + \frac{0.670 Ra_L^{1/4}}{[1 + (0.492/Pr)^{9/16}]^{4/9}}$$



Vertical Plate (cont.)

- The last two correlations presented contain a **constant** (~ 0.68 and ~ 0.825) that takes into account the thermal behavior at small Rayleigh numbers, where the heat conduction through the fluid dominates.
- The two relations are valid for an **isothermal vertical plate**. However, the correlations can be also used for the case of **constant surface heat flux**, as long as the Nusselt and the Rayleigh numbers are estimated using the temperature difference at the plate half-length:

$$\Delta T_{L/2} = T_s \Big|_{L/2} - T_\infty$$

- This is because for a constant surface heat flux the temperature difference becomes a function of the position on the plate.
- It is possible to estimate the $\Delta T_{L/2}$ combining one of the Nusselt number correlation equations with the relation for the heat transfer coefficient, and by solving numerically for $\Delta T_{L/2}$:

$$\overline{Nu}_L = f(\Delta T_{L/2})$$

&

$$\bar{h}_L = \frac{q_s''}{\Delta T_{L/2}}$$



$$\Delta T_{L/2}$$

/ Vertical Cylinders

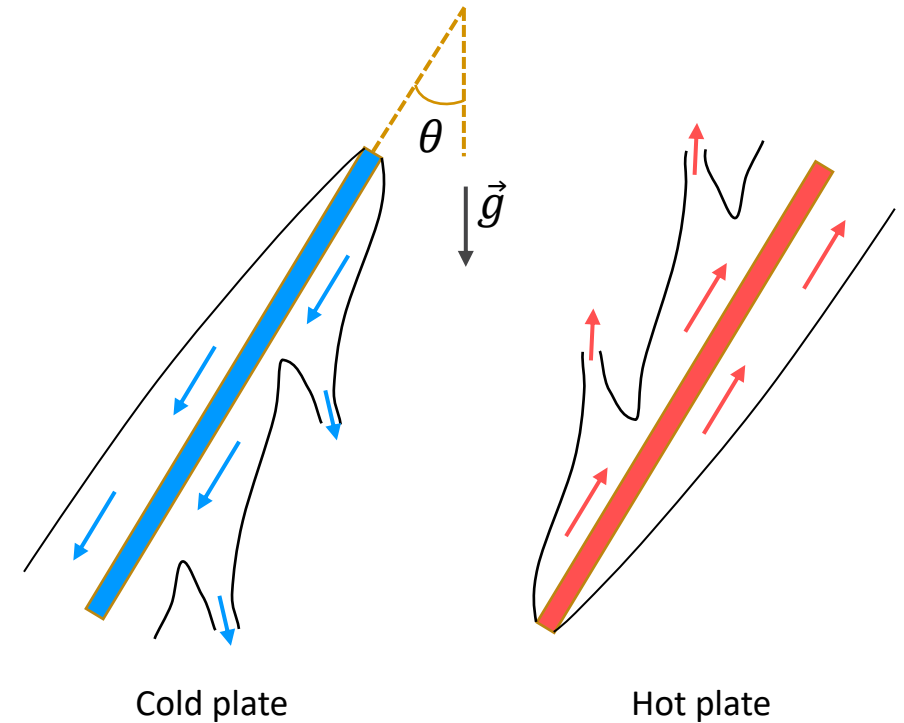
- In the case of vertical cylinders with **diameter much larger than the boundary layer thickness**, we can use the same correlations introduced for the plate.
- If we denote the cylinder height as L and the diameter as D , we can use the following relation to make sure the plate correlations can be applied to the cylinder:

$$\frac{D}{L} \gtrsim \frac{35}{Gr_L^{1/4}}$$

- In the case of slender cylinders, the correlations are no longer valid. Indeed, the curvature affects the way the boundary layer grows and helps to increase the natural convection heat transfer.

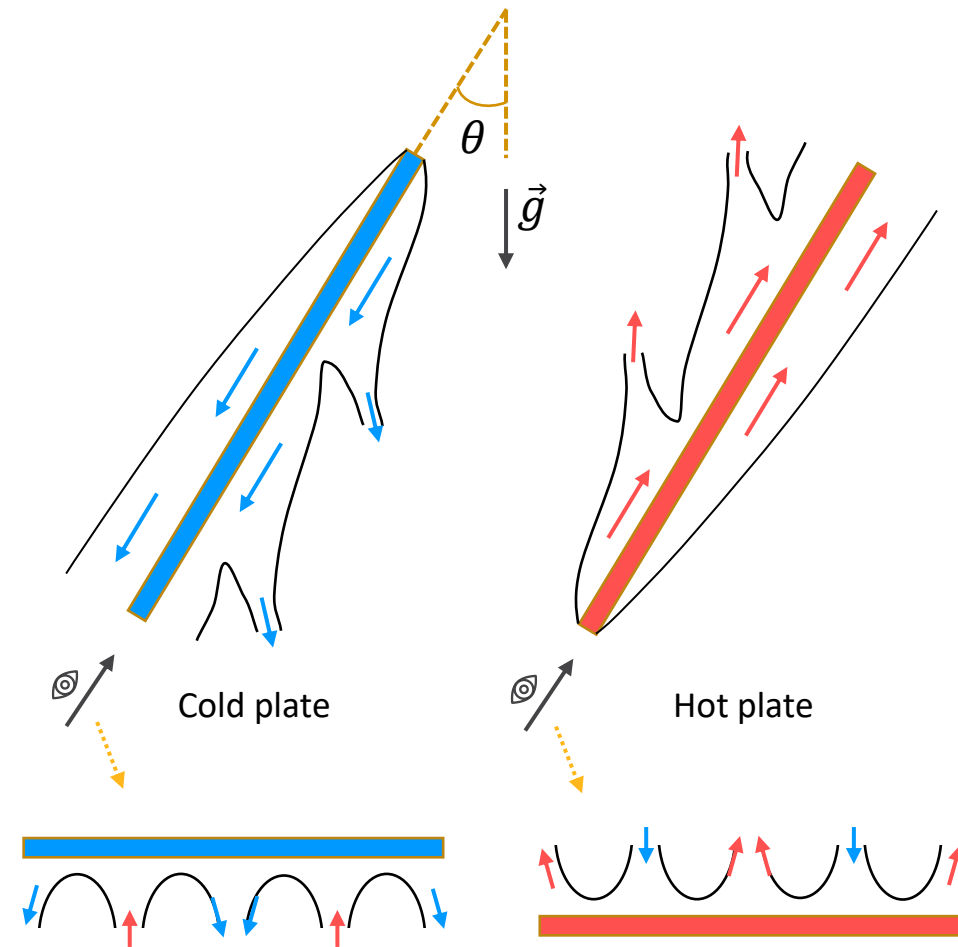
/ Inclined Plate

- A vertical plate is aligned parallel to the gravitational vector. If we rotate the plate by some degrees with respect to the out-of-plane axis, we obtain an inclined plate. The gravitational field would then have a **normal** and a **parallel** component to the plate.
- The inclination would reduce the heat transfer due to convection on the **bottom side of a hot plate** or the **top side of a cold plate**. This is because the parallel component of the gravitational vector is now reduced to $g \cos \theta$, and the convection happens at lower velocities.
- The heat transfer at these sides of hot and cold plates can be estimated using the last two correlations shown before, simply using $g \cos \theta$ (instead of g) for calculating the **Rayleigh number**.
- This approach is valid for inclination angles $0^\circ \leq \theta \lesssim 60^\circ$.



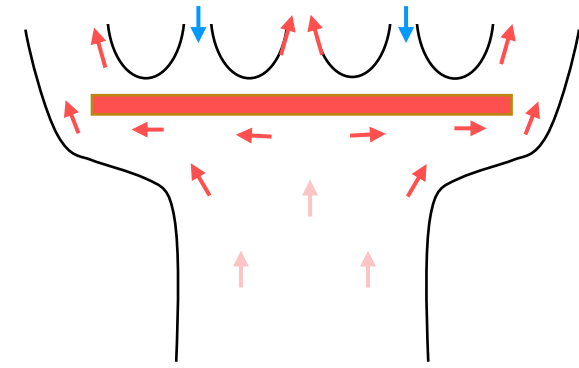
/ Inclined Plate (cont.)

- The **top side of a hot inclined plate** or the **lower side of a cold one** are subject to different phenomena than the opposite sides of the plate.
- The component of the gravitational vector normal to the plate pulls the fluid away from the plate, creating three-dimensional patterns. The boundary layer development is interrupted and the hot (cold) fluid moving away from the hot (cold) plate is replaced by cold (hot) fluid.
- This mechanism **enhances the convection heat transfer** at these sides of the plates.
- Due to the three-dimensional nature of this process and its dependency on the inclination angle, the simple correlations presented before cannot be used to estimate the heat transfer.
- It is possible, however, to find useful correlations in the literature on a case-by-case basis.

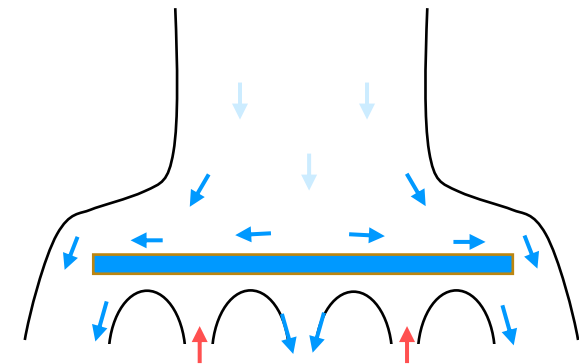


/ Horizontal Plate

- A horizontal plate is **orthogonal to the gravitational field**. The fluid in contact with a hot plate would ascend while the fluid in contact with a cold plate would descend, due to buoyancy.
- On the bottom side of a hot plate or the top side of a cold one, part of the fluid cannot ascend or descend due to the presence of the plate. Hence, it is forced to **move horizontally** along the plate until it reaches the plate ends.
- In this case the heat transfer due to convection is **low** due to the limited fluid motion.
- On the opposite sides of the hot and cold plates, instead, the fluid is free to ascend or descend respectively, moving away from the plate. The fluid is replaced by cold or hot air thanks to buoyancy.
- This three-dimensional motion **significantly increases** the heat transfer of the plate due to natural convection.



Hot plate



Cold plate

Horizontal Plate (cont.)

- The following correlations suggested by Adams can be used to estimate the average Nusselt number.

Hot plate – Upper surface
Cold plate – Lower surface

$$\overline{Nu}_L = 0.54 Ra_L^{1/4}$$

$$10^4 \lesssim Ra_L \lesssim 10^7$$

$$\overline{Nu}_L = 0.15 Ra_L^{1/3}$$

$$10^7 \lesssim Ra_L \lesssim 10^{11}$$

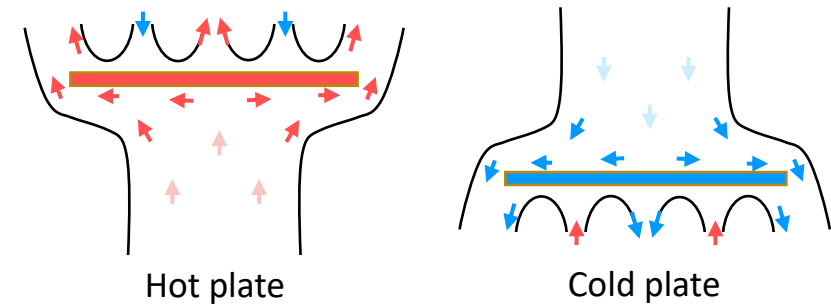
Hot plate – Lower surface
Cold plate – Upper surface

$$\overline{Nu}_L = 0.27 Ra_L^{1/4}$$

$$10^5 \lesssim Ra_L \lesssim 10^{10}$$

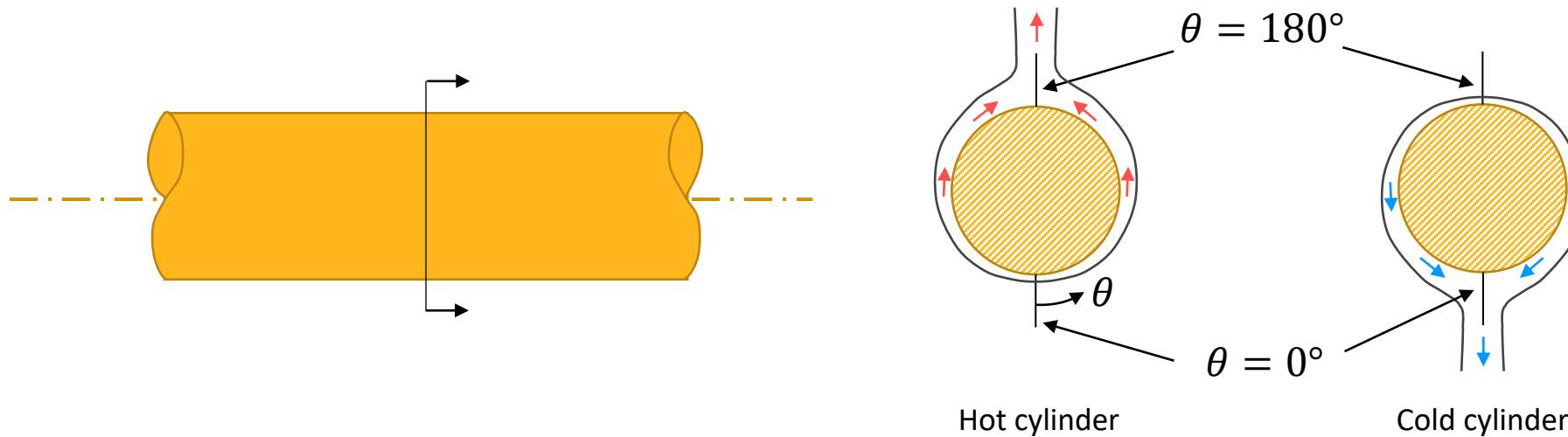
- The accuracy can be improved using a modified characteristic length (L) definition based on the plate surface areas A_s and perimeter P :

$$L \equiv \frac{A_s}{P}$$



Horizontal Cylinder

- External natural convection over long horizontal cylinders considers the heat exchanged by the cylinder with the surrounding fluid.
- For hot cylinder we can see that an ascending plume would form around its surface. The boundary layer starts growing at the bottom of the cylinder ($\theta = 0^\circ$) and near the top ($\theta < 180^\circ$) the fluid separates and continues ascending.
- A similar pattern can be observed around a cold cylinder where the cold fluid would descend starting from the top of the cylinder ($\theta = 180^\circ$) and separate into a plume at $\theta > 0^\circ$.



/ Horizontal Cylinder (cont.)

- We can use different correlations for estimating the Nusselt number averaged along the cylinder circumference. In this case the Rayleigh number would be based on the **cylinder diameter**.
- One correlation is in a simple form where the coefficients vary depending on the range of the Rayleigh number:

$$\overline{Nu}_D = \frac{\bar{h}D}{k} = C Ra_D^n$$

with

Ra_D	C	n
$10^{-10} - 10^{-2}$	0.675	0.058
$10^{-2} - 10^2$	1.020	0.148
$10^2 - 10^4$	0.850	0.188
$10^4 - 10^7$	0.480	0.250
$10^7 - 10^{12}$	0.125	0.333

- Another correlation can be applied as is for a large range of Rayleigh numbers, instead:

$$\overline{Nu}_L = \left\{ 0.60 + \frac{0.387 Ra_D^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2$$

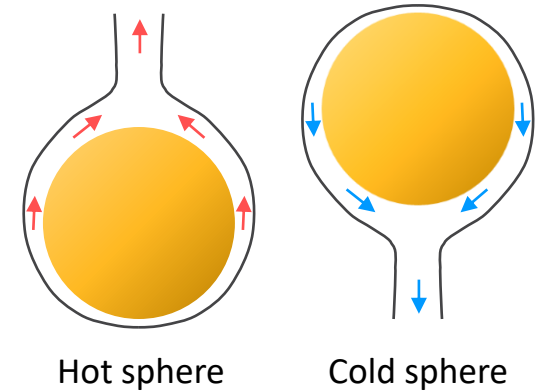
$$Ra_D \lesssim 10^{12}$$

/ Sphere

- Flow patterns around a hot or cold sphere immersed in a static fluid are like the ones seen at the cylinder cross section. The fluid would ascend or descend starting from the bottom or top of the sphere and then separate into an ascending or descending plume near the opposite end of the sphere.
- The following correlation can be used to estimate the average Nusselt number:

$$\overline{Nu}_D = 2 + \frac{0.589 Ra_D^{1/4}}{[1 + (0.469/Pr)^{9/16}]^{4/9}}$$

$$Ra_D \lesssim 10^{11}$$
$$Pr \gtrsim 0.7$$



- It is interesting to note that as the Rayleigh number **approaches zero** the average Nusselt number becomes equal to **2**. This value corresponds to the conduction heat transfer between a sphere and the infinite medium surrounding it.

/ Summary

- In this lesson we analyzed the heat transfer due to external natural convection.
- We discussed in detail different canonical geometries such as plates, cylinders and spheres.
- For each geometry we analyzed the flow pattern due to the natural convection and outlined useful correlations to estimate the average Nusselt number and obtain the average heat transfer coefficient.

 **Ansys**

